

Networking Projects

Search for a project related to network security, network design, architecture, and management.

WANT TO CONTRIBUTE TO THIS PROJECT LIST?

Submit your project idea today. All types of projects are welcomed.

[Submit Your Project Idea →](#)

Why learn networking?

Computer networks act as the foundational system used to the transfer data across any geographical location. Understanding the underlying infrastructure behind how computer networks communicate and function is necessary before adding clearly defined security controls.

Networking Projects Lists

Search for your next network security project idea. All kinds skill levels are included in this following list below.

Click the project title for more details about the project.

NOTE: Cybercademy and its affiliates do not promote the use of the following ideas for unethical and illegal uses, we do not take responsibility for the use of, promotion, or spread of the following ideas with the intent of illegitimate and illegal uses. We assume these project ideas and descriptions are used only for the purpose of at-home learning.

Create a Cybersecurity Homelab

Build a homelab environment simulating different components of a business network. This project can be implemented through virtualization or hardware.

Setup and Manage a SIEM

Setup and manage an open-source or free security information event management (SIEM) platform used to aggregate, organize, and prioritize security alerts.

Configure an IDS & IPS

Configure an open-source or free intrusion detection / prevention system. Once you have completed the initial setup, simulate network traffic and analyze how the system works.

Deep Packet Inspection Project

Deep packet inspection (DPI) is a type of data processing that inspects in detail, data being sent over the network ensuring only authorized information is being transmitted across the network. Use a DPI tool to evaluate how DPI works.

Create Secure Backup Plan

Research different types of backup plans and create your own backup server to store and save important files, folders, and documents.

The Secure File Sharing Service

Configure and use a secure file sharing service, such as SFTP, to securely transfer files over an unsecure medium.

Investigate Open Source Incident Response Tools

Find an open-source incident response tool and configure the program to monitor the network and alert if any potential security events happen.

Create a VPN Server

Simulate a corporate VPN setup within a homelab using OpenVPN.

Setup and Manage an Open Source Firewall

Setup and configure an open source firewall, such as PfSense to filter malicious and unwanted connections.

Setup a Web Server

Create a web server using a web server service such as Apache and add security best practices to the server to ensure the server is hardened.

DNS Server & Security

Research DNS attacks and build a small-scale DNS server. Add the proper security measures to the DNS Server.

Honeypot Testing

A honeypot is a computer security mechanism or “trap” set to detect and deflect unauthorized attempts for proprietary and confidential information. Set up an open source honeypot and try accessing the confidential information.

Nmap Tool & Scanning

Use the popular network scanner and tool, Nmap, to scan and identify hosts and services on the network. Create a small network of vulnerable machines and use Nmap to scan for details and more information on each machine.

Configure a Vulnerability Scanner

Setup and configure a popular vulnerability scanner. Create a network of vulnerable host machines and use the scanner to identify known vulnerabilities.

Create a SIEM with the Open-Source ELK Stack

A SIEM is a technology used to support threat detection, vulnerability management, and security alerts in an organization. Using the open-source ELK stack (Elasticsearch, Logstash, and Kibana), develop security dashboard to trigger and display security alerts when malicious activity is being logged.

